INCH-POUND
MIL-PRF-39035/4G
w/ Amendment 3
8 December 2023
SUPERSEDING
MIL-PRF-39035/4G
w/ Amendment 2
2 June 2017

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PERFORMANCE SPECIFICATION SHEET

RESISTOR, VARIABLE, NONWIREWOUND (ADJUSTMENT TYPE, SINGLE TURN), NONESTABLISHED RELIABILITY, AND ESTABLISHED RELIABILITY STYLE RJR50

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39035.

Characteristic H is inactive for new design.
Use characteristic F as its replacement.

1. SCOPE

1.1 Scope. This specification covers the requirements for style RJR50, nonestablished reliability, and established reliability, adjustment type, single turn, nonwirewound, variable resistors. This style is available in characteristic F. Characteristic H is inactive for new design.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of a basis style of this specification and a coded number. The PIN is in the following form:

RJR50	FP102M
Specification style number	Coded number

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, Ohio 43218-3990 or by email Resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-39035 - Resistor, Variable, Nonwirewound (Adjustment Type), Nonestablished Reliability, and Established Reliability, General Specification for

* Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-39035.

3.2 Interface and physical dimension requirements. Resistors shall meet the interface and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be 1/4 watt for characteristics F and H.

3.4 Terminals. Characteristic F is available in P- type terminal. Characteristic H is inactive for new design.

3.5 Nominal resistance value and maximum rated ac or dc working voltage. Nominal resistance values and maximum rated ac or dc working voltages shall be as specified in table I.

3.5.1 Resistance value deviations. All maximum deviations as specified in this section are to be considered absolute limits with the exception of the contact resistance adjustments.

3.6 Actual effective electrical travel. Actual effective electrical travel shall be 215 degrees, minimum.

3.7 Torque.

3.7.1 Operating. Operating torque shall be a maximum of 2.0 ounces-inches.

3.7.2 Stop. The stop shall withstand a force of 4 ounce-inches when this amount of torque is applied through the operating shaft to the stop.

3.8 Maximum voltage. Maximum rated ac or dc working voltage shall be 200 volts.

3.9 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of resistor components and solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum of 3 percent lead, by mass (see 6.3).

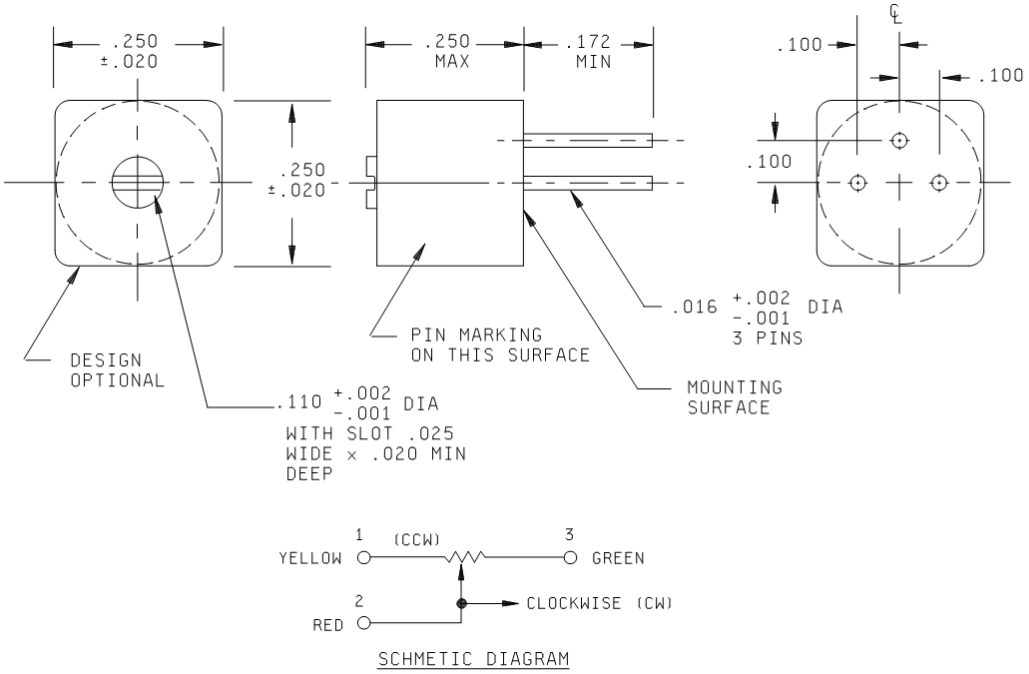
3.10 Marking. Due to size limitations, this style resistor shall be marked with the following minimum information:

7523A	-	Date code, lot code
FP102MJ	-	Characteristic, terminal, resistance value, failure rate and JAN marking
12345	-	Source code

The complete marking is required on the unit package.

* 3.11 Weight. The weight is 0.021 ounces (0.6 grams).

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.001	0.03	0.015	0.38	0.025	0.64	0.172	4.37
0.002	0.05	0.016	0.41	0.100	2.54	0.250	6.35
0.010	0.25	0.020	0.51	0.110	2.79		

NOTES:

1. Dimensions are in inches.
* 2. Unless otherwise specified, tolerance is ±0.005 (0.13 mm).
3. Metric equivalents are given for general information only.
4. Mounting means are by use of pins only.
5. The picturization of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
6. The head of the actuating screw may or may not be flush with or recessed in the body.

FIGURE 1. Style RJR50 resistors.

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TABLE I. Nominal resistance value and maximum rated ac or dc working voltage.

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Nominal resistance value (in Ohms)	Maximum rated ac or dc working voltage per characteristics F, and H 1/	Nominal resistance value (in Ohms)	Maximum rated ac or dc working voltage per characteristics F, and H 1/
10	1.58	20,000	70.7
20	2.23	25,000	79.0
50	3.54	50,000	111
100	5.0		
200	7.07	megohms	
500	11.1	0.10	158
1,000	15.8	0.20	200
2,000	22.3	0.25	200
5,000	35.4	0.50	200
10,000	50.0	1.00	200

1/ Characteristic H are inactive for new design.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-39035.

4.2 Visual and mechanical inspection. Visual and mechanical inspection shall be in accordance with MIL-PRF-39035.

4.3 Dielectric withstanding voltage. The magnitude of test voltage shall be 600 volts rms at atmospheric pressure, and 250 volts rms at reduced barometric pressure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-39035 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and the complete PIN (see 1.2).

b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.

c. Packaging instructions (see 5.1).

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6.3 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.4 Amendment notation. The margins of this specification are marked with an asterisk to indicate modification generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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